Answers

1.**D**	2.**D**	3.**B**	4.**D**	5.**C**	6.**B**	7.**B**	8.**A**
9.**B**	10.**A**	11.**C**	12.**A**	13.**B**	14.**A**	15.**A**	16.**A**
17.**C**	18.**A**	19.**B**	20.**D**	21.**A**	22.**6**	23.**C**	24.**D**
25.**C**	26.**C**	27.**B**	28.**D**	29.**A**	30.**B**	31.**B**	32.**A**
33.**C**	34.**C**	35.**D**	36.**A**	37.**C**	38.**D**	39.**C**	40.**D**
41.**A**	42.**E**	43.**B**	44.**C**	45.**C**	46.**D**	47.**B**	48.**D**
49.**D**	50.**D**	51.**A**	52.**D**	53.**B**	54.**C**	55.**B**	56.**D**
57.**C**	58.**A**	59.**B**	60.**D**	61.**B**	62.**A**	63.**3**	64.**4**
65.**A**	66.**B**	67.**C**	68.**B**	69.**D**	70.**A**	71.**D**	72.**A**
73.**C**	74.**A**	75.**A**	76.**B**	77.**C**	78.**B**	79.**B**	80.**B**
81.**C**	82.**B**	83.**D**	84.**A**	85.**A**	86.**A**	87.**D**	88.**C**
89.**D**	90.**C**	91.**C**	92.**C**	93.**D**	94.**A**	95.**B**	96.**D**
97.**C**	98.**B**	99.**C**	100.**D**	101.**A**	102.**B**	103.**A**	

Explanations

1.**D**

From table it is clear that for University of California - Berkeley annual tuition fee does not exceed $23,000 and median starting salary is at least $70,000. Hence option D.

VIEW SOLUTION

2.**D**

The number of people with age less than 40 are at least:

In no of children 0 category => 1 male and 1 female

In no of children 1 category => 1 male and 1 female

In no of children 2 category => 1 male and 1 female

In no of children 3 category => 2 male and 1 female

There are atleast 9 people who have age less than 40.

9 people out of 30 people => 30%

VIEW SOLUTION

3.**B**

Summation of sales value of all coffees (excluding others) = 31.15 + 26.75 + 15.25 + 17.45 = 90.6

Total value = 132.8

Others = 132.8 - 90.6 = 42.2

% share for others = $\frac{42.2}{132.8} \times 100$ = 31.77 $\approx$ 32

VIEW SOLUTION

4.**D**

From the given data it is clear that Dipan is the only student who obtained Group Scores of at least 95 in every group and obtained the highest Group Score in Social Science Group. Hence option D.

VIEW SOLUTION

5. **C**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B . Now if statement 1 is true then , Firm B is aggressive ltd. and Firm C is Honest Ltd. And also statement 2 is fulfilled.

VIEW SOLUTION

6. **B**

The common interest of all the people in option B is EG.

But, all the four are committed to projects in March => They cannot attend EG.

Hence, people in option 4 cannot attend a workshop.

VIEW SOLUTION

7. **B**

Gayatri and Urvashi are interested in only one workshop each, that is, EG. So they cannot attend more than one workshop.
Zeena is interested in all three workshops but she is committed to internal projects during the months of Jan and Mar. So she cannot attend the CS and EG workshops.
These are 3 executives gayatri , urvashi , zeena who can't attend more than 1 workshop.
Similarly using the same above logic we find that Anshul, Bushkant, Charu, Eshwaran, Fatima, Hari, Indiri, John, Mandeep, Nandlal, Rahul, Sunita and Yamini cannot also attend more than one workshop.
So, in total there are only 3 executives who cannot attent more than one work shop.

VIEW SOLUTION

8. **A**

Calculating the tons produced per hectare of rice cultivation for different states ,we get highest value for punjab and haryana , the ratio being above 5 for both . Hence option A.

VIEW SOLUTION

9. **B**

Value added per employee = Value added / Employment . So we can see that Central and State/local governments have highest Value added per employee = 1.8/1 = 1.8

VIEW SOLUTION

10. **A**

If the total number of factories is 100, then the total number of employees = 60 × 100 = 6000 of which 64.6% = 3876 work in wholly private factories. Since the number of wholly private factories = 90.3, So we have 3876/90.3 = 43. hence option A.

VIEW SOLUTION

11. **C**
Man's new gross income = 8000 + (8000*0.8) + (8000*0.1) = 15200

Average gross income of HR = 5000 + (5000*0.7) = 8500

New average = $\frac{8500*5+15200}{6}$ = 9616

% increase = $\frac{1116}{8500} * 100$ = 13.12% => 13% approximately.

VIEW SOLUTION

12. **A**
Percentage capacity of production for Lipton coffee = 64.8%
Let's say maximum production = x
So 1.64 = $x \times \frac{64.8}{100}$
or $x = 2.53$

VIEW SOLUTION

13. **B**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.
Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by coconuts is $3 * \frac{3}{3+2} = 3 * \frac{3}{5} = 1.8$ lakhs

Cost of a coconut is Rs. 5

Hence, the total output of coconuts is $\frac{180000}{5} = 36,000$

VIEW SOLUTION

14. **A**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by lemons is $3 * \frac{2}{3+2} = 3 * \frac{2}{5} = 1.2$ lakhs

Total land used to cultivate lemons is 5 acres.

So, revenue generated by the lemons per acre is $\frac{1.2}{5} = 0.24$ lakhs per acre

VIEW SOLUTION

15. **A**
For 160 defective pieces:
Using 1st test:
Cost of testing = $1000 \times 2 = 2000$
Cost correcting 80% pieces = $128 \times 25 = 3200$
Cost of penalty on 20% pieces = $32 \times 50 = 1600$

Total = 6800

Using 2nd test:
Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all pieces = $160 \times 25 = 4000$
Total = 7000

Using no test:
Cost of penalty on all pieces = $160 \times 50 = 8000$

Hence, he should use first test only.

VIEW SOLUTION

16. **A**
There are 6 airports from USA which are in top 10 busiest aiports. Hence 600/10 = 60%. Hence option A.

VIEW SOLUTION

17. **C**
Total passengers handled in top 5 airports are 336648008 . So percentage of passengers handled by heathrow airport is (62263710*100)/(336648008) which is approximately equal to 20%.

VIEW SOLUTION

18. **A**
This percentage is maximum in February in J&K; where the percentage is $\frac{11183}{11285} = 99.09$%

VIEW SOLUTION

19. **B**
Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Hence, Maximum withdrawal was after second year.

VIEW SOLUTION

20. **D**
Even if If the prices include a margin of 15% over the total cost that the company incurs, the total cost company incurs would be minimum for route AHJ i.e 2350 km. Hence option D.

VIDEO SOLUTION

21. **A**
Among the top ten students, only Dipan scored at least 95 in at least one paper from each of the groups. Hence option A.

VIEW SOLUTION

22. **6**
The airports with serial numbers 1, 3, 4, 6, 8 and 11 are of type ‘1’ and handle more than 35 lakh passengers.

VIEW SOLUTION

23. **C**
Let's say the principal amount is P, Rate is R.
1st withdrawal = $PR + \frac{P}{5}$
Remaining money = $\frac{4P}{5}$

2nd withdrawal = $(\frac{4PR}{5} + \frac{2P}{5})$
Remaining money = $\frac{2P}{5}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$
Remaining money = $\frac{P}{5}$

4th withdrawal = $\frac{P}{5} + \frac{PR}{5}$ = 11000
Or P = 50000 (Putting R = 10%)

VIEW SOLUTION

24. **D**
Based on Condition **II**, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl missed his Mathematics examination.

Further, based on Condition **III**, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

$\therefore$ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option D: Esha and Foni.

VIDEO SOLUTION

25. **C**

There are atleast 4 respondents that fall into the 35 to 40 years age group (both inclusive) . So 400/30 = 13.333 % . Hence , the percentage of respondents that fall into the 35 to 40 years age group (both inclusive) is at least 13.333%.

VIEW SOLUTION

Data Interpretation for CAT Questions (download pdf)

26. **C**

Let the age of an employee which were mutual transfer between Marketing and Finance departments be x(M to F) and y also age of one employee transfeered from Marketing to HR be z.

As a result, the average age of finance department increased by one year , so we have (30*20-x+y)/20 = 31 and that of Marketing department remained the same so we have (35*30+y-z-x)/29 = 35.

Solving both we have z = 15.

So new average age of HR department is (45*5 + 15)/6 = 40

Hence option C.

VIEW SOLUTION

27. **B**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option B: Alva and Deep.

VIDEO SOLUTION

28. **D**

10 lakh ton = 1 million ton. States whose annual rice production per million of population is at least 400,000 tons are Haryana , Gujarat, Punjab , Madhya pradesh, Tamil nadu, maharashtra, uttar pradesh,andhra pradesh . Hence 8 states.

VIEW SOLUTION

29. **A**

If A acquires firm B , the total sales of 2 firms is 50036 . now the percent market share can be given by 50036*100/89576 which is equal to around 55% . Hence option A.

VIEW SOLUTION

30. **B**

Maximum difference in ranks between Japan and India is 4 which is highest when compared to other countries. hence option B.

VIEW SOLUTION

Logical Reasoning for CAT Questions (download pdf)

31. **B**

We know that Compound productivity = Gross output / Fixed capital .So compound productivity for Public sector = 0.6, Central Government = 0.725, States/Local = 0.47, Central/States/Local = 1.07, Joint sector = 1.23 and wholly private = 1.36. Hence required is Wholly private, joint, central and state/local.

VIEW SOLUTION

32. **A**

The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

Date	Order Placed	1-day Delivery	2-day Delivery	Lost	Delivery done on the date
11			4		
12	14	7	6	1	
13	31	21	8	2	11
14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6=1=14.

Fraction of orders booked on 15th that were lost = 12/28

Fraction of orders booked on 16th that were lost = 2/25

Fraction of orders booked on 13th that were lost =2/31

Fraction of orders booked on 14th that were lost = 8/30.

∴ Option A is correct answer.

VIDEO SOLUTION

33. **C**

Let's total coffee production is = x

So $x \times \frac{61.3}{100} = 11.6$

x = 18.9 (nearly)

VIEW SOLUTION

34. **C**

The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

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14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

FRom the table we can determine that among options, number of orders booked on 13th are maximum.

Average time can be calculated as follows

	x	y	x+2y	x+y	(x+2y)/(x+y)
13	21	8	37	29	1.275862069
14	15	3	21	18	1.166666667
15	8	8	24	16	1.5
16	13	10	33	23	1.434782609

14 is the least

VIDEO SOLUTION

35. **D**
The maximum difference of the ranks between all traits is largest for Japan and Thailand which is 4. Hence option D.

VIEW SOLUTION

Quantitative Aptitude for CAT Questions (download pdf)

36. **A**
Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Maximum Interest was after the first year (i.e. PR)

VIEW SOLUTION

37. **C**
The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

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14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

The total number of orders placed on 13th = 21+8+2 = 31

FRom the table we can determine that among options, number of orders booked on 13th are maximum.

VIDEO SOLUTION

38. **D**
From the table we can see that , profitability in A , B ,C , D (4914)/(24568) , (4075)/(25468) , (4750)/(23752) , (3946)/(15782) . So clearly D has the highest profitability.

VIEW SOLUTION

39. **C**
There are atmost 23 people with age greater than 35 . Hence (23/30)*100 = 76.67 . So option C.

VIEW SOLUTION

40. **D**
We know that number of factories in joint sector is 1.8% = 2700, so number of factories in Central Government = 1% of (2700 x 100/1.8) = 1500.

Value added by Central Government = 14.1% of 14,000= Rs. 1974 crores.

So we get = (1974 crores)/(1500) = Rs. 1.32 crore

VIEW SOLUTION

Know the CAT Percentile Required for IIM Calls

41. **A**
Let total rural population in China be A and urban population in China be B.

Total population in India = T = A + B

Based on the sanitation facilities table,

$\frac{74}{100}$ B + $\frac{7}{100}$ A = $\frac{24}{100}$(A + B)

Solving, we get

$\frac{A}{B}$ = $\frac{50}{17}$

A : B = 50 : 17

Rural population % = $\frac{50}{50+17} \times 100$ = $\frac{50}{67} \times 100$ = 74.6%

Let total rural population in Indonesia be C and urban population in Indonesia be D.

Total population in India = T = C + D

Based on the sanitation facilities table,

$\frac{40}{100}$ C + $\frac{73}{100}$ D = $\frac{51}{100}$(C + D)

Solving, we get

$\frac{C}{D}$ = $\frac{22}{11}$

C : D = 2 : 1

Rural population % = $\frac{2}{2+1} \times 100$ = $\frac{2}{3} \times 100$ = 66.7%

Let total rural population in Philippines be R and urban population in Philippines be U.

Total population in India = T = R + U

Based on the sanitation facilities table,

$\frac{88}{100}$ U + $\frac{66}{100}$ R = $\frac{77}{100}$(U + R)

Solving, we get

$\frac{U}{R}$ = $\frac{11}{11}$

U : R = 1 : 1

Rural population % = $\frac{1}{1+1} \times 100$ = $\frac{1}{2} \times 100$ = 50%

Ascending order: Philippines, Indonesia, China

Therefore, answer is option A.

VIEW SOLUTION

42. **E**

Lets first consider ayesha's marks , her 7 marks will increase in geography . So increase in her average marks is 7/(2*5) = 0.7. So new average is 96.9 . </br>
Now for dipan , his 5 marks will increase in mathematics . So increase in his average 5/5 = 1 . So his new average is 96+1 = 97 which is greater than ayesha's . Other will clearly have a lower average than these both. Hence option E.

VIEW SOLUTION

43. **B**

There are only 2 colleges - New York university and stanford university which are uniformly better (higher median starting salary AND lower tuition fee) than Dartmouth College.

VIEW SOLUTION

44. **C**

The truck should be sent on a day on which maximum number of units can be sent. It happens on 2nd, 4th, 5th and 7th day. Hence the answer is b.

VIEW SOLUTION

45. **C**

We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

By Statement 1, Truthful is A or C.

If Profitable's lowest revenue are from UP, Profitable is either A or D.

By Statement 2, Honest and Aggressive are either A-D or B-C. As Profitable is either A or D, Honest and Aggressive will be one of B or C.

Hence, Truthful is A => Profitable is D.

Hence, Truthful's lowest revenues are from UP.

VIEW SOLUTION

46. **D**

The success rate of moving from written test to interview stage for males in 2003 can be given by 637/60133 and for females in the same year can be given by 399/40763 . We can see that the rate of males is clearly more that that of females. hence statement A is false.

Now the success rate of moving from written test to interview stage for females in 2002 was 138/15389 and for females in the year 2003 was 399/40763.So the rate in 2002 is less than that in 2003. . Hence both statements are false.

VIEW SOLUTION

47. **B**

With no incomplete innings and BA of 50 , lets assume $r_1 = 50$ and $n_1 = 1$.

Now we have $BA = MBA_2 = 50$

So we have $r_2 = 45$ and $n_2 = 1$.

We have $BA = 95$ and $MBA_2 = \frac{95}{2} < 50$.

Hence MBA_2 decreases.

So BA will increase and MBA_2 will decrease.

VIEW SOLUTION

48. **D**

Lets take $n_1 = n_2 = 10$ and $r_1 = r_2 = 100$.

Using given formula we have $BA = 20$ and $MBA_1 = MBA_2 = 10$.

So we have BA highest which is not the case in option A,B and C. Hence the correct option is D.

VIEW SOLUTION

49. **D**

Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st withdrawal = $PR + \frac{P}{5} = \frac{15P}{50}$

2nd withdrawal = $\frac{4PR}{5} + \frac{2P}{5}$ = $\frac{24P}{50}$

3rd withdrawal = $\frac{2PR}{5} + \frac{P}{5}$ = $\frac{12P}{50}$

4th withdrawal = $\frac{PR}{5} + \frac{P}{5}$ = $\frac{11P}{50}$

Lowest withdrawal was after the fourth year.

VIEW SOLUTION

50. **D**

Manually counting lays which are are used to produce XL yellow or XL white fabrics. We get 15 number of such lays.

VIEW SOLUTION

51. **A**

Manually counting lays for which number of lays produced is non-zero . We get 15 lays. Hence option A.

VIEW SOLUTION

52. **D**

Manually counting non-zero value number of fabrics in yellow fabric category. We get 14. Hence option D.

1, 3, 4, 6, 7, 8, 9, 11, 12, 15, 21, 24, 25, 27

VIEW SOLUTION

53. **B**

To calculate the number of trees required, let us calculate the total number of tonnes supplied.

The total number of tonnes supplied equals 494,525.

One tree supplies 40 kilograms.

Hence, the number of trees required is $\frac{494525*1000}{40} \approx 12.5$ million trees

VIEW SOLUTION

54. **C**
From the table it is clear that the state which supplied the most number of apples is either HP or J&K.; UP and Cold Storage supplied less than a thousand apples in total.

The total number of apples supplied by HP is 231,028
The total number of apples supplied by J&K; is 262,735

Hence, the correct answer is J&K;

VIEW SOLUTION

55. **B**
Only 4 states Haryana , Punjab , Maharashtra, Andhra Pradesh have a per capita production of rice greater than Gujarat. Hence option B.

VIEW SOLUTION

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56. **D**
The maximum difference in ranks in any of the parameters is termed to be dissimilarity. Difference between the ranks of Malaysia and China in Vision (V) = 5-1 =4 Difference between the ranks of China and Thailand in Vision (V) = 5-1 = 4 Difference between the ranks of Thailand and Japan in Decisiveness (D) = 5 -1 =4 The maximum rank difference between Japan and Malaysia is in Vision (V) and Negotiation skills (N). The maximum difference in rank is 4. Therefore, option D is the odd one.

VIEW SOLUTION

57. **C**
The least disparity => The least difference between the coverage in urban drinking water facilities and the coverage in urban sanitation facilities.

Phillipines has a disparity of 92-88 = 4, which is the least.

Hence, option C is the answer.

VIEW SOLUTION

58. **A**
When there are 200 defective pieces:
Using test 1
Cost of testing= 2000
Cost of correcting 80% pieces = 4000
Cost of penalty on 20% pieces = 2000
Total = 8000

Using test 2
Cost of testing = 3000
Cost of correcting all pieces = 5000

Total = 8000

Using no test
Cost of penalty on all pieces= 10000

Hence he can use either 1st test or second test.

VIEW SOLUTION

59. **B**
The minimum cost from A to J we know is 2275.

Let the CP to company be C

Since 10% over actual CP is the total price i.e. $\mathrm{CP} \times 1.1 = 2275 \rightarrow CP = \frac{2275}{1.1}$

The total distance is 1950+1400=2350 Km.

Cost per Km = $\frac{\frac{2275}{1.1}}{2350}$ = Rs 0.88/Km

VIDEO SOLUTION

60. **D**
The cumulative orders booked by 19th are 337 and that of 18th are 332=> No. orders booked on 19th are 5

Similarly we can find the orders booked on that day till 14th.

Number of orders lost that were booked on 12th = Cumulative orders lost till 14th-Cumulative orders lost till 13th =92-91=1

Similarly, the number of orders lost till 17th can be found out.

Number of orders delivered on 13th are 11 out of which 4 are orders which were booked in 11th so, 7 must be the orders which were booked on 12th.

Similarly, we can find the orders which took 1day and 2 days to get delivered till 17th.

Date	Order Placed	1-day Delivery	2-day Delivery	Lost	Delivery done on the date
11			4		
12	14	7	6	1	
13	31	21	8	2	11
14	30	15	3	12	27
15	28	8	8	12	23
16	25	13	10	2	11
17	25	3	13	9	21
18	5	1			13
19	5				14

Now, total number of orders booked on 12th will be 7+6+1=14.

From the table we can determine that among options, number of orders booked on 13th are maximum.

For 15 the delivery ratio = 8/8 = 1

For 16 the delivery ratio = 13/10 = 1.3

For 13 the delivery ratio = 21/8 = 2.625

For 14 the delivery ratio = 15/3 = 5

Hence Option D

VIDEO SOLUTION

61. **B**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores.

For Alva: best 3 out of 4 - 80(English), 75(Hindi), 75(Science)

Avg. = 230/3 = 76.67 $\neq$ 70

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl most likely missed his Mathematics examination.

For Foni: best 3 out of 4 - 83(English), 83(Social Science), 88(Science)

Avg. = 254/3 = 84.67 $\neq$ 78

Hence, we observe that only Carl has missed his Mathematics examination. Hence, Option B is the correct answer.

VIDEO SOLUTION

62. **A**

The price per kilogram sales of each of the four brands is given below.

Brooke Bond - Rs. 122.16
Nestle - Rs. 131.77
Lipton - Rs. 121.03
MAC - Rs. 118.71

Hence, the correct answer is Nestle

VIEW SOLUTION

63. **3**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

$\therefore$ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is **3**. {Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi)}

VIDEO SOLUTION

64. **4**

Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Except for Alva and Deep, we can conclusively comment of the missed subjects of the rest four. Hence, the correct answer is **4**.

VIDEO SOLUTION

65. **A**

From the table we can see that, the lowest price would be from A to H and H to J.

The cost of travel from A to H = Rs 1850

The cost of travel from H to J = Rs 425

Total cost = 1850 + 425 = Rs 2275.

VIDEO SOLUTION

66. **B**

From the table we can see that, the lowest price would be from A to H and H to J.

The cost of travel from A to H = Rs 1850

The cost of travel from H to J = Rs 425

Total cost = 1850 + 425 = Rs 2275

Lowest price = Rs 2275

95% of 2275 = Rs 2161

VIDEO SOLUTION

67. **C**

If the airports C, D and H are closed down the minimum price to be paid by a passenger travelling from A to J would be by first travelling to F and then from F to J.

The cost of travel from A to F = Rs 1700

The cost of travel from F to J = Rs 1150

Total cost = 1700 + 1150 = Rs 2850

VIDEO SOLUTION

68. **B**

The new average basic pay of HR dept. would be (5000*5 + 6000*2 + 8000)/8 = 45000/8.

So % change in basic pay is [(45000/8) - 5000]/5000 = 12.5%.

VIEW SOLUTION

69. **D**

In 2002, the number of females selected for the course as a proportion of the number of females who bought application forms was 48/19236 and for males was 171/61205. So rate for males was higher than that of female. Hence option A is false.

In 2002, among those called for interview, the success rate for females was 48/138 and for males 171/684. So the rate was higher for females .Both are false. Hence option D.

VIEW SOLUTION

70. **A**

The percentage of absentees in the written test among females in 2002 was 3847/19236 and in 2003 was 4529/45292. Thus percentage of absentees decreased from 02 to 03. Hence statement A is true.

The percentage of absentees in the written test among males in 2003 is 3065/63298 which is clearly less than that of female in 2003. Hence statement b is false . Hence option A.

VIEW SOLUTION

71. **D**
We can clearly make out from the given table that there are 8 schools in the list which have single digit rankings on at least 3 of the 4 parameters.

VIEW SOLUTION

72. **A**
If we find maximum of difference in ranks for all traits between India and others we find that the difference is least for India and china with just 2 points. Hence china is least dissimilar to India.

VIEW SOLUTION

73. **C**
Since we know the average , we can calculate the marks Dipan got in English Paper II (x) as ; 98+95+95.5+95+(96+x)/2 = 96*5 . We get 96+x = 96.5*2 . Thus x = 97 . Hence option C .

VIEW SOLUTION

74. **A**
Had Joseph, Agni, Pritam and Tirna each obtained Group Score of 100 in the Social Science Group , their final score would increase by 0.9, 0.9, 2.2, 2.1 respectively. Adding these to the final score then their standing in decreasing order of final score would be Pritam, Joseph, Tirna, Agni with scores of 96.1,95.9,95.8,95.2 respectively. Hence option A.

VIEW SOLUTION

75. **A**
Since we require atleast 2 women and 1 women is selected , other girl can be any one out of the yamini or lavanya. Now seeing the options and given data , we can infer that dinesh and anshul cannot attend as they have projects in january and rahul can attend it. So we have rahul and yamini. Hence option A.

VIEW SOLUTION

76. **B**
We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B .
Now if Profitable Ltd. has the lowest share in MP market then Firm B is profitable ltd. Honest ltd. would be any firm out of A and D. So now if statement 1 is true then statement 2 is false as honest will have more revenue than profitable. hence option c.

VIEW SOLUTION

77. **C**
We have ,

	FIRM A	FIRM B	FIRM C	FIRM D
Total revenue	190	217	222	185

There is a difference of 5 million between Firm A and Firm D and also between Firm C and Firm B . Now if statement 1 is true then Honest ltd. would be firm B , and aggressive ltd. would be firm C. But now if we consider statement 2 we get Agressive ltd. as firm B and Honest ltd as firm C. Thus any 1 out of the 2 statements can be true and not both together. hence option C.

VIEW SOLUTION

78. **B**
Counting the number of variety of fabrics which have positive non-zero surplus, we get 4 such varieties. Hence option B.

VIEW SOLUTION

79. **B**
1 Million = 10 lakhs. Using the conversion and manually counting the international airports of type 'A' accounting for more than 40 million passengers, we get 5 such airports. Hence option B .

VIEW SOLUTION

80. **B**
All the top 20 busiest airports handle more than 30 million passengers. So, we have to just count the airports which are not located in the USA, out of these 20 airports. The airports ranked 4, 6, 7, 8, 11, 18 are located outside the USA. Thus, there are 6 such airports.
Hence, option B is the correct answer.

VIEW SOLUTION

81. **C**
We can find Value added/employment and value added/fixed capital for the different sectors. We get values as Wholly private 0.9 and 1.25; Joint sector 1.59 and 1.19; Central/State/Local 1.8, 1.28; others 0.92 and 0.75. Hence option C.

VIEW SOLUTION

82. **B**
There are 5 countries which lie from 10 Deg E to 40 Deg E , but out of those only 1 lie in southern hemisphere. hence 1*100/5 = 20%.

VIEW SOLUTION

83. **D**
There are 11 cities whose names begin with a consonant and are in the Northern Hemisphere in the table and there are 13 number of countries whose name begins with a consonant and are in the east of the meridian. Hence option D.

VIEW SOLUTION

84. **A**

Number of countries whose name starts with vowels and located in the southern hemisphere are 3 and number of countries, the name of whose capital cities starts with a vowel in the table is 2.

Hence the ratio 3:2.

VIEW SOLUTION

85. **A**
Let's say the principal amount is P, Rate is R (i.e. R = 10%)

1st Interest = PR

2nd Interest = $\frac{4PR}{5}$

3rd Interest = $\frac{2PR}{5}$

4th Interest = $\frac{PR}{5}$

Total interest = $\frac{12PR}{5} = 12000$

VIEW SOLUTION

86. **A**
We should consider three possible cases as 1. when he is using test 1
2. When he is using test 2
3. When he is using no test
Now we will choose a method where the total expenditure will be least.
So for option A, let's consider that number of defective pieces are 50.
Hence, while using test 1:
Cost of testing = $1000 \times 2 = 2000$
Correcting 80% of pieces = $40 \times 25 = 1000$
Penalty for 20% of pieces = $10 \times 50 = 500$
Total = 3500

For test 2:
Cost of testing = $1000 \times 3 = 3000$
Correcting all 50 pieces = $50 \times 25 = 1250$
Total= 4250

For no test:
Penalty for all 50 pieces = $50 \times 50 = 2500$

Hence cost is least when he is using no test while number of defective pieces is less than 100.

VIEW SOLUTION

87. **D**
When there are 120 widgets defective:

Using 1st test:
Cost of testing = $1000 \times 2 = 2000$
Cost of correcting 80% pieces = $96 \times 25 = 2400$
Cost of penalty on 20% pieces = $24 \times 50 = 1200$
Total cost = 5600

Using 2nd test:

Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all 120 pieces = $120 \times 25 = 3000$
Total cost = 6000

Using no test:

Total cost = 120 x 50 = 6000

Hence, he should use 1st test only.

VIEW SOLUTION

88. **C**
Let's say defective pieces are 300
Using test 1
Cost of testing= $1000 \times 2 = 2000$
Cost of correcting 80% pieces = $240 \times 25 = 6000$
Cost of penalty on 20% pieces = $60 \times 50 = 3000$
Total cost = 11000

Using test 2
Cost of testing = $1000 \times 3 = 3000$
Cost of correcting all 300 pieces = $300 \times 25 = 7500$
Total cost = 10500

Using no test:
Cost of penalty on all pieces = $300 \times 50 = 15000$

Hence, he should use 2nd test.

VIEW SOLUTION

89. **D**
Maximum unutilized capacity => Minimum unitized capacity

MAC has the least utilized capacity.

VIEW SOLUTION

90. **C**
If the truck is hired only for the 7th day, the cost will be (150*6 + 180*5 + 120*4 + 250*3 + 160*2 + 120)*0.8 = 2776
So, total cost incurred = 2776 + 1000 = 3776

If a truck is hired on 4th day and 7th day, the cost incurred will be (150*3 + 180*2 + 120)*0.8 + 1000 + (160*2 + 120)*0.8 + 1000 = 3096

So, a truck should be hired on the 4th and 7th day.

VIEW SOLUTION

91. **C**
The answer to this question is the same as the state which supplied the maximum number of apples.

We can rule out UP and Cold Storage as they supplied significantly less than the other two.
Among the other two states,

The total number of apples supplied by HP is 231,028
The total number of apples supplied by J&K; is 262,735

Hence, the correct answer is J&K;

VIEW SOLUTION

92. **C**
The supply is greater than or equal to demand if the cold storage number is zero.

This is the case between the months May-September.

Hence, the correct option is option (c)

VIEW SOLUTION

93. **D**
In the earlier question, we determined that we needed 12.36 million trees per year.
Each hectare has 250 trees.

So, number of hectares of land used is $\frac{12.36*1000000}{250} \approx 49450$

VIEW SOLUTION

94. **A**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
As the revenue is split equally, Gopal's share in the revenue is $3 * \frac{1}{2} = 1.5$ lakhs

VIEW SOLUTION

95. **B**
Let the amounts paid by Gopal and Ram to Krishna be 2X and 3X.

Gopal invested 2 lakhs additionally.
So, the total amount spent by him is 2X+2.
As the total amounts spent by the two of them are equal, it implies that 2X+2 = 3X or X = 2
Hence, the cost of the land is 2X+3X = 5X = 10 lakhs.
Total investment by the two of them is 10+2 = 12 lakhs

Hence, the total revenue generated is 12*25% = 3 lakhs.
The revenue generated by coconuts is $3 * \frac{3}{3+2} = 3 * \frac{3}{5} = 1.8$ lakhs

The number of coconut trees is 5 times the number of lemon trees = 100*5 = 500
So, the value of output per trees for coconuts is $\frac{180000}{500} = 360$

VIEW SOLUTION

96. **D**
We know the total revenue generated by coconuts and lemons respectively.
We also know the price of a coconut and can thus calculate the total number of coconuts produced per acre.
We don't know the price of a lemon and thus can't calculate the total number of lemons produced per acre.

So, the answer can't be determined.

VIEW SOLUTION

97. **C**
Bangladesh has the highest percentage in drinking water facilities. Let us see which countries it dominates. This is possible if it has greater sanitation facilities also as compared to the other country.

Bangladesh has better sanitation facilities than India, China, Pakistan and Nepal. So those Bangladesh dominates those four countries and they are not on the coverage frontier.

Similarly, Philippines has the highest percentage of sanitation facilities. Let us see which countries it dominates. This is possible if it has greater drinking water coverage also.

Philippines has greater drinking water coverage than all countries except Bangladesh. Hence, it dominates all the countries except Bangladesh. So, no other country is on the coverage frontier.

Hence, the only countries on the coverage frontier are Bangladesh and Philippines.

VIEW SOLUTION

98. **B**
India has lesser sanitation facilities than Pakistan. Hence it is false that India > Pakistan. So statement A is false.

India has greater sanitation facilities as well as drinking water coverage than China. Similar is the case between India and Nepal.
Hence, India > China and India > Nepal. So, statement B is true.

Sri Lanka has lesser drinking water coverage than China. Hence, statement C is false.

China has greater sanitation facilities as well as drinking water coverage than Nepal. Hence, China > Nepal and statement D is true.

So, the only two true statements are B and D.

VIEW SOLUTION

99. **C**
Let total rural population in India be R and urban population in India be U.

Total population in India = T = U + R

Based on the sanitation facilities table,

$\frac{70}{100}$ U + $\frac{14}{100}$ R = $\frac{29}{100}$(U + R)

Solving, we get

$\frac{U}{R}$ = $\frac{15}{41}$

U : R = 15 : 41

Rural population % = $\frac{41}{15+41} \times 100$ = $\frac{41}{56} \times 100$ = 73.2%

VIEW SOLUTION

100. **D**
For a country to be on the coverage frontier, it should not be dominated by any other country. That is no other country should have both (sanitation facilities and drinking water coverage) better than it.
The first three options, though true, are not the reason for India not being on the coverage frontier. That is because they don't mention the other parameter at all.

For example, option A doesn't mention sanitation facilites while options B and C don't mention drinking water coverage. Hence, none of A,B and C are the reason India is not on the coverage frontier.

Option D is factually wrong as Indonesia has lesser drinking water coverage than India and doesn't dominate it.

Hence, none of the options given are correct.

VIEW SOLUTION

101. **A**
The most disparity in rural sector is in India = 79-14 = 65

The disparity in all other countries is less than 65.

VIEW SOLUTION

102. **B**
The first 9 airports handle more than 40 lakh passengers each, out of which 3 are in AP and 6 are outside AP.

VIEW SOLUTION

103. **A**
Based on Condition II, we understand that the student who missed the Mathematics examination did not miss any other examination. This indicates that the Maths score is bound to be the average of the best 3 out of the 4 exam scores obtained by this candidate. Based on this inference, we can proceed with identifying the math score that can be represented as an average of the rest of the scores. We can straightaway eliminate Deep and Esha as potential candidates, given that their Mathematics score is greater than the rest of the exam scores. After estimating the average scores for the rest of the candidates, we observe that only Carl has missed his Mathematics examination.

For Carl: best 3 out of 4 - 80(Hindi), 90(Social Science), 100(Science)

Avg. = 270/3 = 90 which matches the given value

∴ Carl missed his Mathematics examination.

Further, based on Condition III, we can surmise that the student who missed Hindi and Science should have similar average scores in these two subjects. We notice that Alva has the same score of 75 in both Hindi and Science. The same can be said about Deep, who has a score of 90 in both these subjects. Thus, one out of Alva and Deep missed out on Hindi and Science examination, while the second individual missed out only on the Hindi examination.

Since we know that Carl, Alva and Deep are unlikely to have missed out on the English exam, we can divert our attention to determining which individual out of Bithi, Esha and Foni failed to appear for this subject. However, we notice that Bithi's English score is greater than the rest of her scores, thereby helping us eliminate her as the potential candidate.

For Esha: best 3 out of 4 - 85(Hindi), 95(Mathematics), 60(Science)

Avg. = 240/3 = 80 which matches the given value

∴ Esha most likely missed her English examination.

For Foni: best 3 out of 4 - 78(Mathematics), 83(Social Science), 88(Science)

Avg. = 249/3 = 83 which matches the given value

∴ Foni most likely missed her English examination.

Based on Condition I, we know that exactly two candidates missed the examinations for English, Hindi, Science, and Social Science.

For English, we determined these individuals to be Esha and Foni. For Hindi, we determined these individuals to be Alva and Deep. For Science, we know one of the individuals is either Alva or Deep. Given that Carl, Alva and Deep cannot be a part of the group that missed Science or Social Science exam, we can proceed by carefully scrutinizing the rest of the group that includes Bithi, Esha and Foni.

We notice that Bithi has a similar score in both Science and Social Science examination. Assuming that she did miss these exams, let us proceed to check if this was actually the case.

For Bithi: Best 2 out 3 - 90(English), 80(Hindi)

Avg = 170/2 = 85 which matches the given value

∴ Bithi is likely to have missed her Science and Social Science examinations.

We additionally notice that Foni has a similar score in English and Social Science. On considering the best 2 out of 3 scores, the average value of the score for both the subject holds (equal to 83). Thus, we can conclude that Bithi and Foni missed their Social Science examination.

Thus, the students who missed just one exam were: Carl (Mathematics); Esha (English) and one out of Alva and Deep (Hindi).

Hence of the six students, we can correctly determine the missed subjects for four of them (except Alva and Deep):

Mathematics: Carl ; **English**: Esha & Foni ; **Hindi**: Alva & Deep; **Science**: Bithi & one out of Alva and Deep ; **Social Science**: Foni & Bithi

Hence, the correct answer to this question is Option A: Bithi & one out of Alva and Deep.

VIDEO SOLUTION